

# Zihao Zhang, Ph.D.

Designer · Educator · Scholar in Landscape Architecture

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## Summary

Zihao Zhang studies landscape as a product of technology and labor through both theory and making, asking what machines, tools, and unseen work of care produce the environments we call natural. His work critiques the cybernetic legacy in design, which, grounded in Western conceptions of nature, treats environments as systems to be sensed, modeled, and controlled, and proposes a practice that begins where the model ends, among more-than-human agents. His current research turns to Chinese thought and landscapes as a distinct cosmology of technology, nature, and cultivation.

## Higher Education

**Ph.D.** in the Constructed Environment, University of Virginia, School of Architecture, 2021

Dissertation: *Cybernetic Environment: Uncontrollability and Non-communication for a Future of Coexistence*

Committee: Michael Lee, Nana Last, Bradley Cantrell, Michael Gorman

**Master of Landscape Architecture**, University of Virginia, School of Architecture, 2016

**Bachelor of Engineering** in Landscape Architecture, Beijing Forestry University, 2013

## Experience

### Leadership

|           |                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------|
| 2024–2026 | Director, Master of Landscape Architecture Program, Spitzer School of Architecture, CCNY         |
| 2023–2024 | Interim Director, Master of Landscape Architecture Program, Spitzer School of Architecture, CCNY |
| 2023–2025 | Co-chair, ASLA-NY Climate Action Committee                                                       |

### Academic Appointments

|              |                                                                              |
|--------------|------------------------------------------------------------------------------|
| 2026–present | Associate Professor (with tenure; early promotion), City College of New York |
| 2021–2026    | Assistant Professor (tenure-track), City College of New York                 |
| 2019–2021    | Lecturer, Landscape Architecture, University of Virginia                     |
| 2017–2019    | Teaching Fellow, Landscape Architecture, University of Virginia              |
| 2015, 2016   | Graduate Instructor, Summer Design Institute, University of Virginia         |

### Professional Experience

|              |                                                                                             |
|--------------|---------------------------------------------------------------------------------------------|
| 2023–present | Co-founder, OSE: Office of Speculative Ecologies                                            |
| 2014         | Landscape Design Intern, Waterstreet Studio, Charlottesville, VA                            |
| 2013         | Landscape Design Intern, Tsinghua Urban Planning and Design Institute, Beijing              |
| 2012         | Architectural and Urban Design Intern, China Construction Engineering Design Group, Beijing |

## Academic and Professional Honors

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| June 2024  | Journal of Digital Landscape Architecture Award for Scientific Excellence, Vienna   |
| May 2023   | Journal of Digital Landscape Architecture Award for Scientific Excellence, Bernburg |
| April 2019 | Finalist, University Graduate Teaching Awards, University of Virginia               |
| June 2017  | Journal of Digital Landscape Architecture Award for Scientific Excellence, Bernburg |
| 2014       | VA-ASLA Award for Community Service                                                 |

## Research, Scholarship, and Creative Work

### Books

(Under contract) Zhang, Zihao, and Shurui Zhang. *Solarpunk Landscapes: Towards Radical Sustainability*. New York: Routledge.

Zhang, Zihao. *Cybernetics and the Constructed Environment: Design Between Nature and Technology*. New York: Routledge, 2025. <https://doi.org/10.4324/9781003320852>

### Book Chapters

(Forthcoming) Zhang, Zihao. "Do Machines Care? Posthuman Labor in the Design Professions." In *The Routledge Companion to Artificial Intelligence in Architecture*, 2nd ed., edited by Imdat As. New York: Routledge, 2027.

(Forthcoming) Zhang, Zihao, and Shurui Zhang. "Escaping Western Gaze: Chinese Diasporic Urbanism in Flushing." In *Becoming Chinatown: An Anthology of Interiors*, edited by Leyuan Li and Tairan An. Berlin: Jovis, 2027.

(Forthcoming) Zhang, Zihao, and Shurui Zhang. "One Prompt, Job Done: A Syndrome." In *Representing Landscapes: AI, Machine Vision, and Climate Futures*, edited by Nadia Amoroso. New York: Routledge, 2027.

Macwillie, Elizabeth, Shawn Rickenbacker, and Zihao Zhang. "Strengthening Local Ecologies of Care Through Art and Activism in Harlem, NYC." In *Activism in Design Education: Health, Equity, and Climate Action*, edited by Claire Latane and Jean Yang, 95–108. New York: Routledge, 2027. <https://doi.org/10.4324/9781003666073-10>

Zhang, Zihao, and Shurui Zhang. "Before the After: Representing Climate Actions in the Age of AI." In *Representing Landscapes: Visualizing Climate Action*, edited by Nadia Amoroso. Routledge, 2024. <https://doi.org/10.4324/9781003404798> (Peer-reviewed; Major contribution)

Cantrell, Bradley, Zihao Zhang, and Xun Liu. "Artificial intelligence and machine learning in landscape architecture." In *The Routledge Companion to Artificial Intelligence in Architecture*, edited by Imdat As & Prithwish Basu. Taylor & Francis, 2021. (Peer-reviewed; Equal contribution)

Gorman, Michael E., Zihao Zhang, et al. "Collaboration Among Apparently Incommensurable Expertises: A Case Study of Combining Expertises and Perspectives to Manage Climate Change in Coastal Virginia." In *The Third Wave in Science and Technology Studies: Future Research Directions on Expertise and Experience*, 255–72. Springer, 2019. (Peer-reviewed; Major contribution)

### Journal Articles

Fernberg, Phillip, and Zihao Zhang. "Problematizing AI Omnipresence in Landscape Architecture." *Journal of Digital Landscape Architecture* 9 (2024): 746–55. <https://doi.org/10.14627/537752069> (Double-blind peer-reviewed)

Zhang, Zihao, et al. "Robots in the Garden: Artificial Intelligence and Adaptive Landscapes." *Journal of Digital Landscape Architecture* 8 (2023): 264–72. (Double-blind peer-reviewed; Lead author)

- Zhang, Zihao**, and Shurui Zhang. "Mapping Landscape Architects' Expertise in Climate Adaptation with Design Research Projects Over the Past Two Decades." *Landscape Architecture Frontiers* 10.2 (2022): 71–81. (Double-blind peer-reviewed; Lead author)
- Zhang, Zihao**. "Post-Digital Landscape and Post-Digital Culture." *Journal of Digital Landscape Architecture* 7 (2022): 26–35. (Double-blind peer-reviewed)
- Zhang, Zihao**, and Bradley Cantrell. "Cultivated Wildness: Towards Technodiversity and Wildness in Machines." *Landscape Architecture Frontiers* 9.1 (2021): 52–65. (Peer-reviewed; Lead author)
- Zhang, Zihao**, and Xun Liu. "Control and Uncertainty: Towards a Paradigm of Prototyping." *Landscape Architecture Frontiers* 8.4 (2020): 10–25. (Double-blind peer-reviewed; Lead author)
- Zhang, Zihao**. "Cybernetic Environment: A Historical Reflection on System, Design, and Machine Intelligence." *Journal of Digital Landscape Architecture* 5 (2020). (Double-blind peer-reviewed)
- Zhang, Zihao**, and Ben Bowes. "The Future of Artificial Intelligence (AI) and Machine Learning (ML) in Landscape Design: A Case Study in Coastal Virginia, USA." *Journal of Digital Landscape Architecture* 4 (2019): 2–9. (Double-blind peer-reviewed; Lead author)
- Cantrell, Bradley, and **Zihao Zhang**. "A Third Intelligence." *Landscape Architecture Frontiers* 6.2 (2018): 42–51. (Peer-reviewed; Equal authorship)
- Zhang, Zihao**, Andrew Mondschein, and Mona El Khafif. "Human-Tool Assemblage: Designers in the Big Data World." *Journal of Digital Landscape Architecture* 3 (2018): 397–405. (Double-blind peer-reviewed; Lead author)
- Zhang, Zihao**. "Human Factors in Responsive Landscapes: Importance and Method." *Journal of Digital Landscape Architecture* 2 (2017): 10–17. (Double-blind peer-reviewed)

### Manuscripts Under Review

- Zhang, Zihao**. "Jugaad Epistemology: Community Sensing and the Knowledge That Lives in the Gap." *Landscape Architecture Frontiers*, special issue on Sustainable Landscape Architecture Supported by Pioneering Sensing Technology. Under review, 2026.

### Conference Proceedings

- Zhang, Zihao**. "On Laboratories and Machines: Interfacing with Landscape Processes." *CELA 2023 Annual Conference Proceedings*. (Double-blind peer-reviewed)
- Karkut, Nicholas, Alexey Kiriluk, **Zihao Zhang**, Zhigang Zhu. "Vegetation Coverage and Urban Amenity Mapping Using Computer Vision and Machine Learning." *International Conference on Image Processing and Vision Engineering*, 2023. (Double-blind peer-reviewed)
- Chen, G.Q., N. Kashyap, **Z. Zhang**, et al. "Predicting Building Energy Efficiency with Multi-Task Learning and Classification on Multimedia Data." *International Conference on SMART MULTIMEDIA*, 2024. (Double-blind peer-reviewed)
- Cantrell, Bradley, and **Zihao Zhang**. "Choreographing Intelligent Agents." *107th ACSA Annual Meeting Proceedings*, 2018.

### Book Reviews

- Zhang, Zihao**. "The Dawn of Everything by David Graeber and David Wengrow." *Journal of Landscape Architecture* 2–2024: 88–89. <https://doi.org/10.1080/18626033.2024.2466341>

## Essays and Op-eds

Zhang, Shurui, and **Zihao Zhang**. "Resilient Schist: Crafting Harlem's Corridor of Art, Culture, and Ecology." *New York Amsterdam News*, January 30, 2025.

**Zhang, Zihao**. "AI in Landscape Architecture: Beyond the Myth of AI-Human Rivalry." *Landezine*, January 6, 2025.  
<https://landezine.com/ai-in-landscape-architecture-beyond-the-myth-of-ai-human-rivalry/>

## Selected Presentations and Talks

### 2025

- "Do Machines Care? Another Solarpunk Manifesto." University of Massachusetts Amherst. October 2, 2025
- Invited Panelist. Digital Landscape Architecture Conference, Dessau (virtual). June 4, 2025
- "Do Machines Care?" Beyond Maintenance Colloquium, ETH Zürich. February 28, 2025

### 2024

- Invited lecture. University of Colorado Boulder. September 10, 2024
- Invited lecture. "Of Bicycles and Artificial Intelligence." CCNY/CWE. March 25, 2024
- Invited lecture. University of Pennsylvania Stuart Weitzman School of Design. February 28, 2024

### 2023

- Invited panelist. ASLA 2023 Annual Conference, Minneapolis. October 29, 2023
- Invited webinar. Landscape Architecture Foundation. August 29, 2023 (*1000+ live attendees, LAF record*)
- Invited panelist. Bi-City Biennale of Urbanism\Architecture (UABB), Shenzhen. January 7, 2023

### 2018–2022 (selected)

- "Post-Digital Landscape and Post-Digital Culture." Digital Landscape Architecture Conference, Harvard GSD, 2022
- "Cybernetic Environment." Digital Landscape Architecture Conference (virtual), 2020
- Invited respondent for N. Katherine Hayles' lecture, UVA, 2020
- Digital Landscape Architecture Conferences: Dessau 2019, Freising 2018, Bernburg 2017

## Other Scholarly and Creative Work

|              |                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023–present | <b>Laboratory for Living Systems</b> , NYC. Co-director                                                                                                                                                       |
| 2022         | <b>Ecological Laboratory for Urban Agriculture (ELUA)</b> , NYC. Designer, PI. <a href="https://ce-lab.one/ELUA">https://ce-lab.one/ELUA</a>                                                                  |
| 2020         | <b>Ostenda illuminata</b> , Public installation, Charlottesville, VA. Research associate: conceptual design, software development, and early prototyping (with Mona El Khafif, Andrew Mondschein, Eric Field) |
| 2018         | <b>CyborgBloom</b> , Digital Art Installation, Charlottesville, VA                                                                                                                                            |
| 2018         | " <b>Forming Atmosphere</b> ", Landscape design research featured in <i>Codify</i> (Routledge)                                                                                                                |

## Public Scholarship

|              |                                                                                                                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–present | <b>Chubby Cat Happy Chat Podcast</b> . Co-host with Shurui Zhang. Mandarin Chinese podcast on design, philosophy, art, and technology. <a href="https://chubbycathappychat.podbean.com/">https://chubbycathappychat.podbean.com/</a> |
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## Grants, Fellowships, and Awards

### Grants

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|-----------|--------------------------------------------------------------------------------------------|
| 2024–2027 | <b>DOE Climate Lighthouse</b> . Co-PI. \$1,000,000.                                        |
| 2024–2025 | <b>Architectural League Independent Projects Grant</b> . PI (with Shurui Zhang). \$10,000. |

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|-----------|-------------------------------------------------------------------------------------------------------|
| 2024      | <b>CUNY GRTI Round 24.</b> PI. \$9,538.67.                                                            |
| 2022–2025 | <b>CCNY Energizing Equity.</b> Co-PI. \$600,000. ( <i>ended</i> )                                     |
| 2022–2023 | <b>CUNY Interdisciplinary Research Grant, Synergistic Agriculture.</b> PI. \$40,000. ( <i>ended</i> ) |
| 2023–2024 | <b>PSC-CUNY, Robots in the Garden.</b> PI. \$6,000. ( <i>ended</i> )                                  |
| 2022–2023 | <b>PSC-CUNY, Rooftop Thermodynamics.</b> PI. \$6,000. ( <i>ended</i> )                                |
| 2022      | <b>Spitzer Faculty Book Project Award.</b> \$4,000.                                                   |

## Fellowships

|           |                                                                                   |
|-----------|-----------------------------------------------------------------------------------|
| 2023–2024 | CUNY Career Success Fellow                                                        |
| 2022–2023 | Deans' Equity and Inclusion Initiative (DEII) Fellow, Dumbarton Oaks              |
| 2020–2021 | Graduate Teaching Fellow (\$24,000), University of Virginia                       |
| 2017–2018 | Digital Art Fellow, University of Virginia                                        |
| 2016–2019 | PhD Scholarship with full tuition + stipend (\$25,000/yr), University of Virginia |

## Professional Activities

### Boards, Workshops, and Committees

|              |                                                                    |
|--------------|--------------------------------------------------------------------|
| 2025–present | Executive Editorial Board, <i>Landscape Architecture Frontiers</i> |
| 2024–present | Advisory Board, West Harlem Art Fund                               |
| 2023–2026    | Board member (ex officio), ASLA-NY                                 |
| 2023–present | Editorial Board, <i>Journal of Digital Landscape Architecture</i>  |
| 2023–2024    | Editorial Board, <i>PLOS ONE</i>                                   |

### Funding Proposal Reviewer

|      |                                                          |
|------|----------------------------------------------------------|
| 2025 | BARD, US-Israel Agricultural Research & Development Fund |
| 2024 | Austrian Science Fund                                    |
| 2023 | European Commission's European Research Council          |

### Consulting

|           |                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------|
| 2023–2025 | Landscape consultant, Masaryk Towers (Mitchell-Lama affordable housing), Lower East Side, NYC |
|-----------|-----------------------------------------------------------------------------------------------|

### Membership

|              |                                                            |
|--------------|------------------------------------------------------------|
| 2021–present | American Society of Landscape Architects, New York Chapter |
|--------------|------------------------------------------------------------|

## Instructional Activities

### City College of New York

|             |                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Fall 2026   | LAAR 61400, Representation and Computation I (3 cr)                                                                                    |
| Spring 2026 | LAAR 66100, Research Studio 2, <i>Designing Urban Pedogenesis: Soil Infrastructures in NYC</i> (Co-instructor with Luke Harris) (6 cr) |
| Spring 2026 | LAAR 65500, Research Workshop (3 cr)                                                                                                   |
| Fall 2025   | LAAR 63100/ARCH 51000, <i>Solarpunk: Scripting Flushing Bay</i> (6 cr)                                                                 |

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| Fall 2024   | LAAR 63100/65100, <i>Technological Sublime in a Watery City</i> (6 cr)   |
| Spring 2024 | SUS 7501, <i>Climate Justice in NYC</i> (1 cr)                           |
| Fall 2023   | LAAR 63100/65100, <i>Technologic</i> (6 cr)                              |
| Spring 2023 | ARCH/LAAR Topics, <i>Machines and Ecologies</i> (3 cr)                   |
| Fall 2022   | LAAR 61100, <i>Contaminated Diversity</i> (6 cr)                         |
| Spring 2022 | LAAR 62100, <i>Cultivated Wildness: Decentralize Central Park</i> (6 cr) |
| Fall 2021   | LAAR 61100, <i>Cultivated Wildness: Decentralize Central Park</i> (6 cr) |
| Fall 2021   | LAAR 61400, Digital and Traditional Drawing (3 cr)                       |

### University of Virginia

|             |                                                        |
|-------------|--------------------------------------------------------|
| Fall 2020   | SARC/LAR6710, Design Computation (Co-instructor)       |
| Fall 2019   | LAR6010, MLA Foundation Studio (Section instructor)    |
| Spring 2019 | LAR7020, Foundation Studio 4 (Teaching Fellow)         |
| Spring 2018 | LAR 8020, Landscape Manifesto Studio (Teaching Fellow) |

## University and Departmental Service

### Committees

|              |                                                                                                |
|--------------|------------------------------------------------------------------------------------------------|
| 2025–present | Executive Committee, Sustainability in the Urban Environment Master Program                    |
| 2023–present | Personnel and Budget; Curriculum; Graduate Programs Steering Committees, Dept. of Architecture |
| 2022–present | Course and Standing Committee, Dept. of Architecture                                           |
| 2022–2023    | Faculty Secretary, Dept. of Architecture                                                       |
| 2021–present | Graduate Program Admission Committee, Dept. of Architecture                                    |

### Search Committees

|           |                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| 2026      | Chair, search committee for Associate/Full Professor of Architecture and Urban AI, Dept. of Architecture                    |
| 2026      | Member, search committees for Lecturer, Sustainability MUE Program; Lab Technician, Robotics Lab; Assistant Professor of LA |
| 2024      | Member, search committees for Professor and Director of MLA Program; Assistant Professor of LA                              |
| 2022–2023 | Member, search committee for Assistant Director, Bond Center for Urban Futures                                              |

## Community Outreach

|              |                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|
| 2024–present | West Harlem Art Fund and Brotherhood Sister Sol, climate resilience advocacy                           |
| 2023         | ELUA and SSA rooftop garden tours; partnerships with NYC Composting Project and CCNY Rangel Initiative |
| 2021–2022    | GrowNYC and NYC-GreenThumb community outreach as part of Synergistic Agriculture project               |
| 2018         | Community-centered Urban Sensing outreach, Charlottesville, VA                                         |

## Graduate Research Experience

### University of Virginia, School of Architecture

- 2018 MainStreet21 – Research Coordination Network (PI: Ila Berman, NSF)
- 2015–2016 Winneba Lagoon project, Ghana (PI: Nancy Takahashi)
- 2013–2016 Lhasa Virtual Reality Project (PI: David Germano)
- 2013–2014 James River Viewshed Studies (PI: Guoping Huang)

### University of Virginia, School of Engineering

- 2018–2021 dMIST – Data-driven Management for Interdependent Stormwater and Transportation Systems (PI: Jonathan Goodall, NSF)